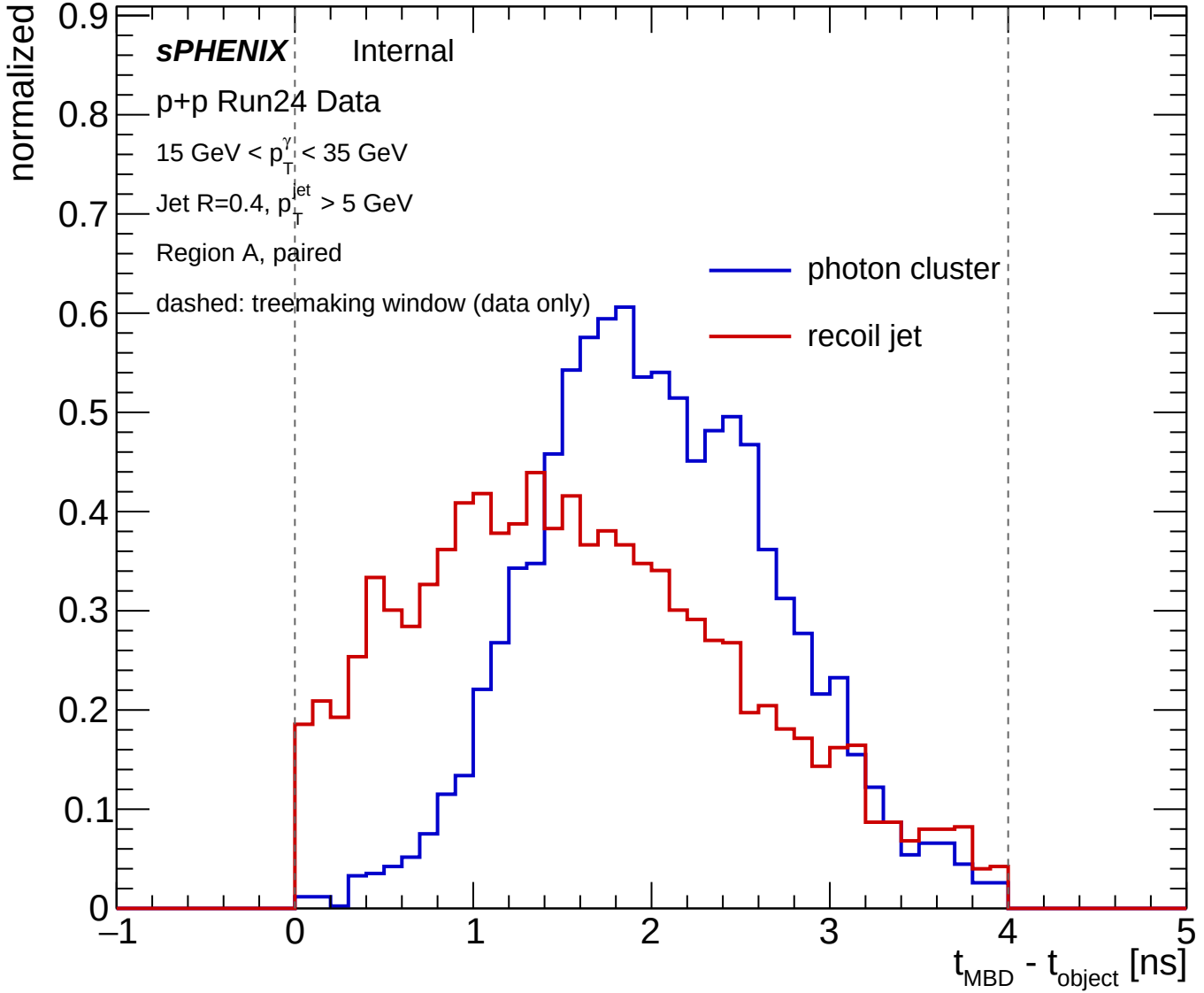
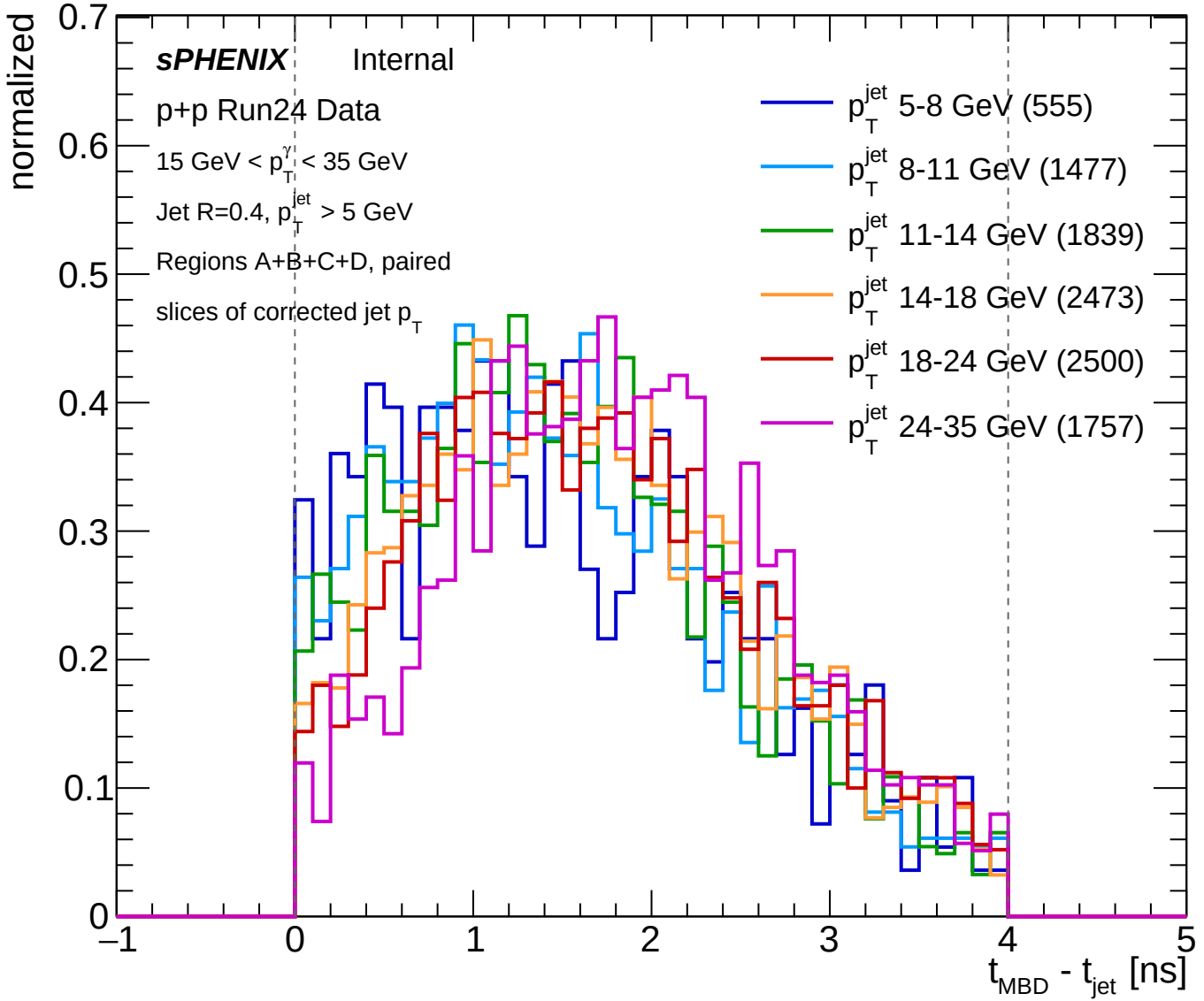
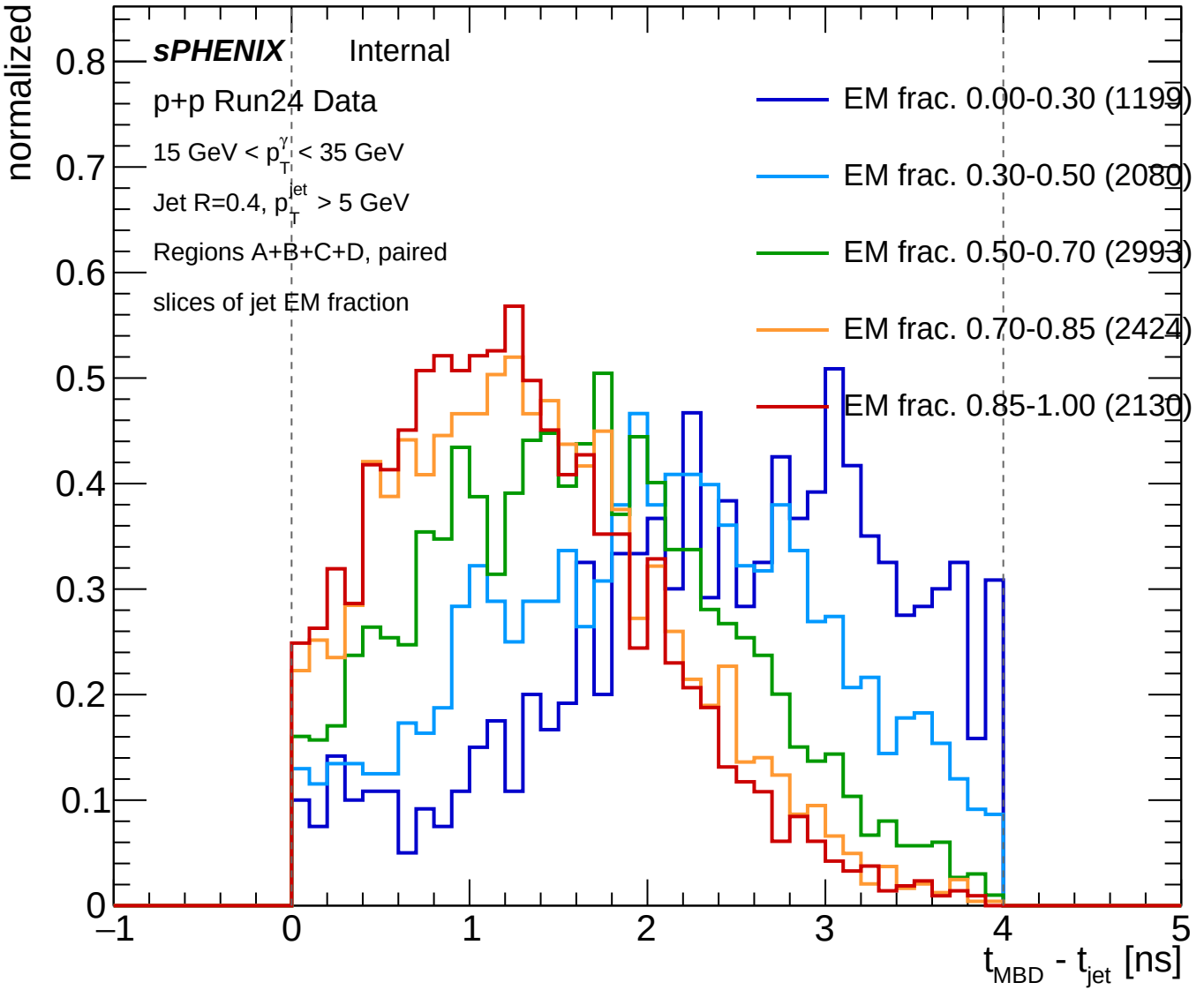
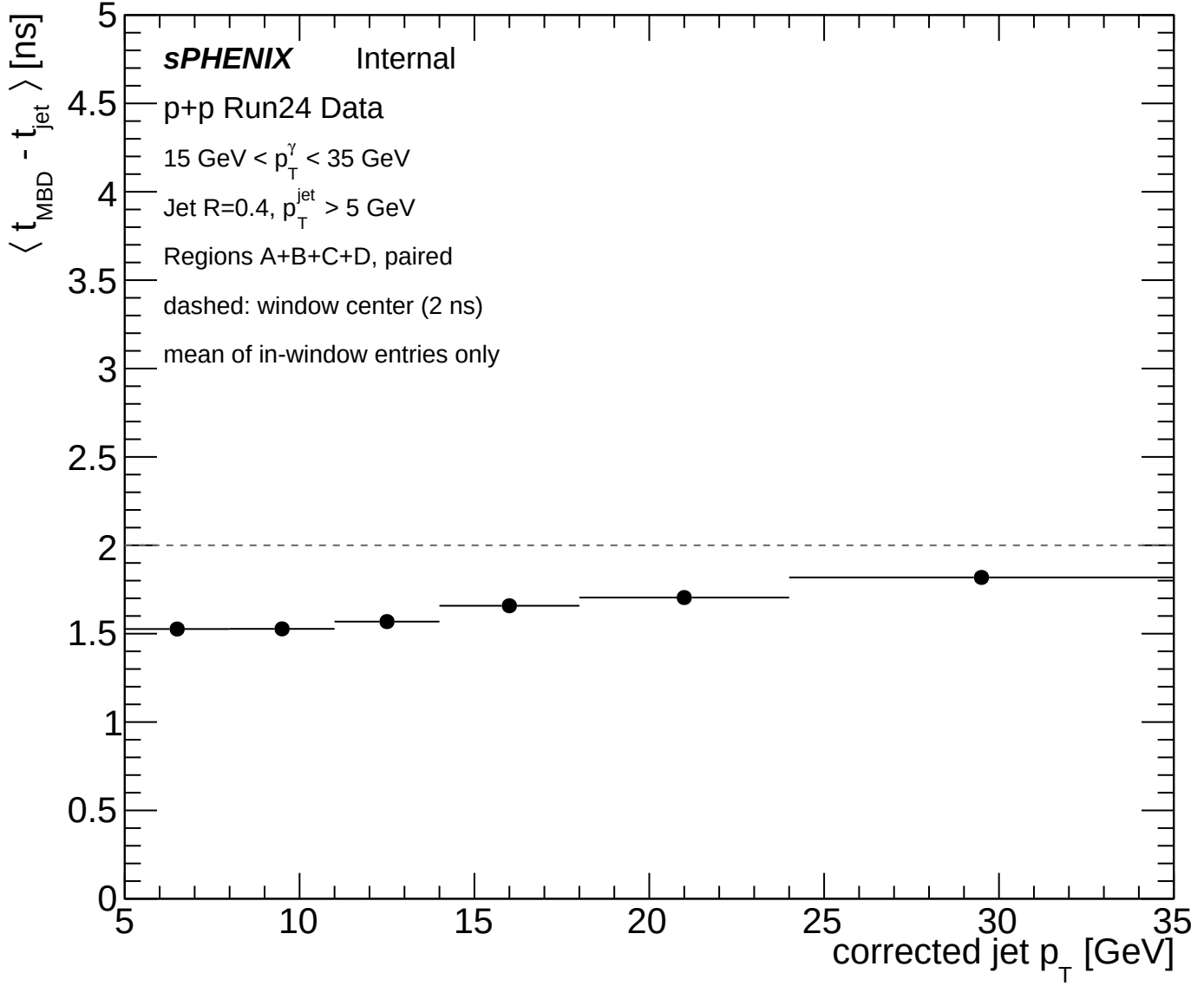
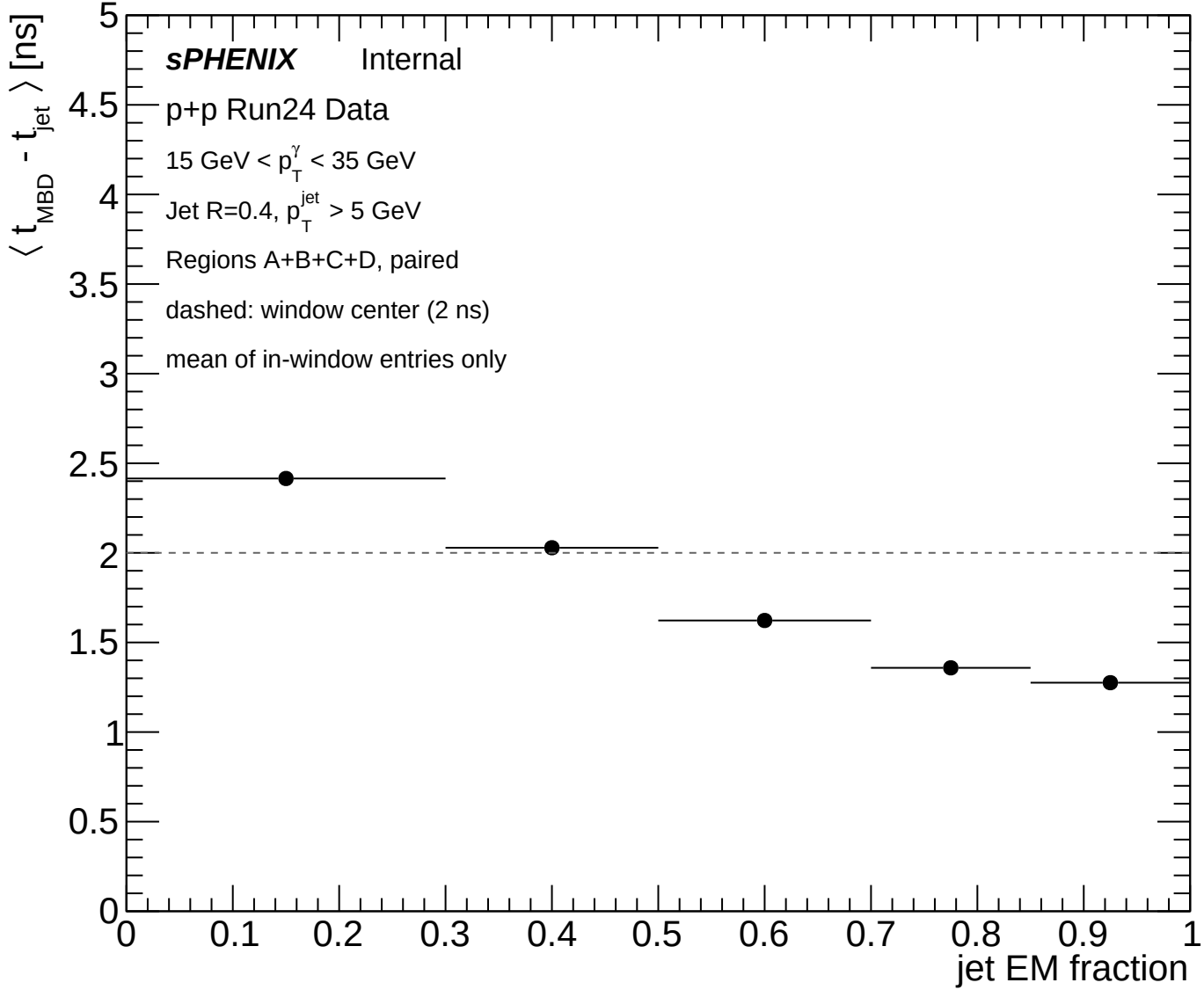
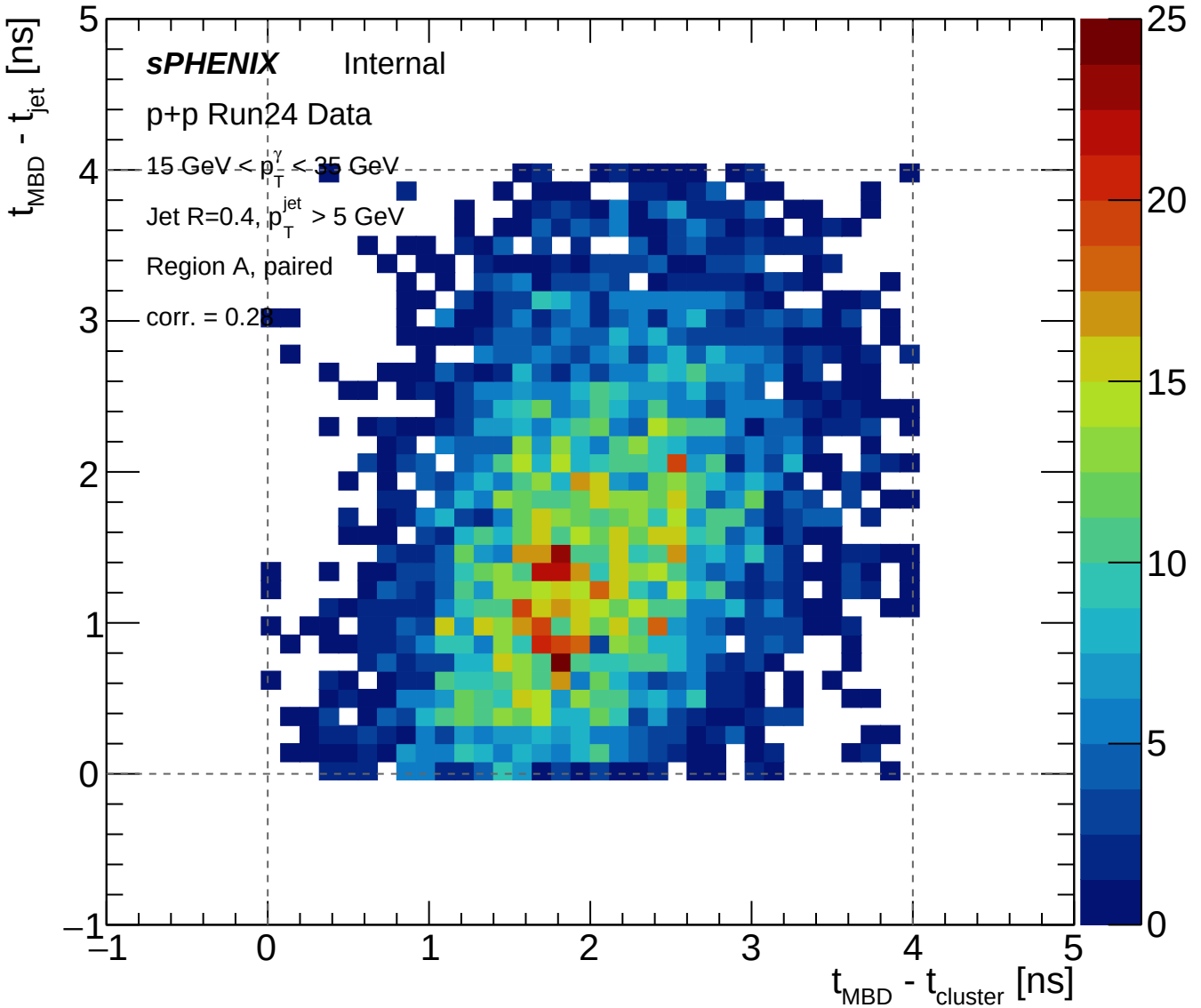

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Region A, paired
dashed: treemaking window (data only)
photon cluster
recoil jet
normalized
t_{MBD} - t_{object} [ns]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
−1
0
1
2
3
4
5

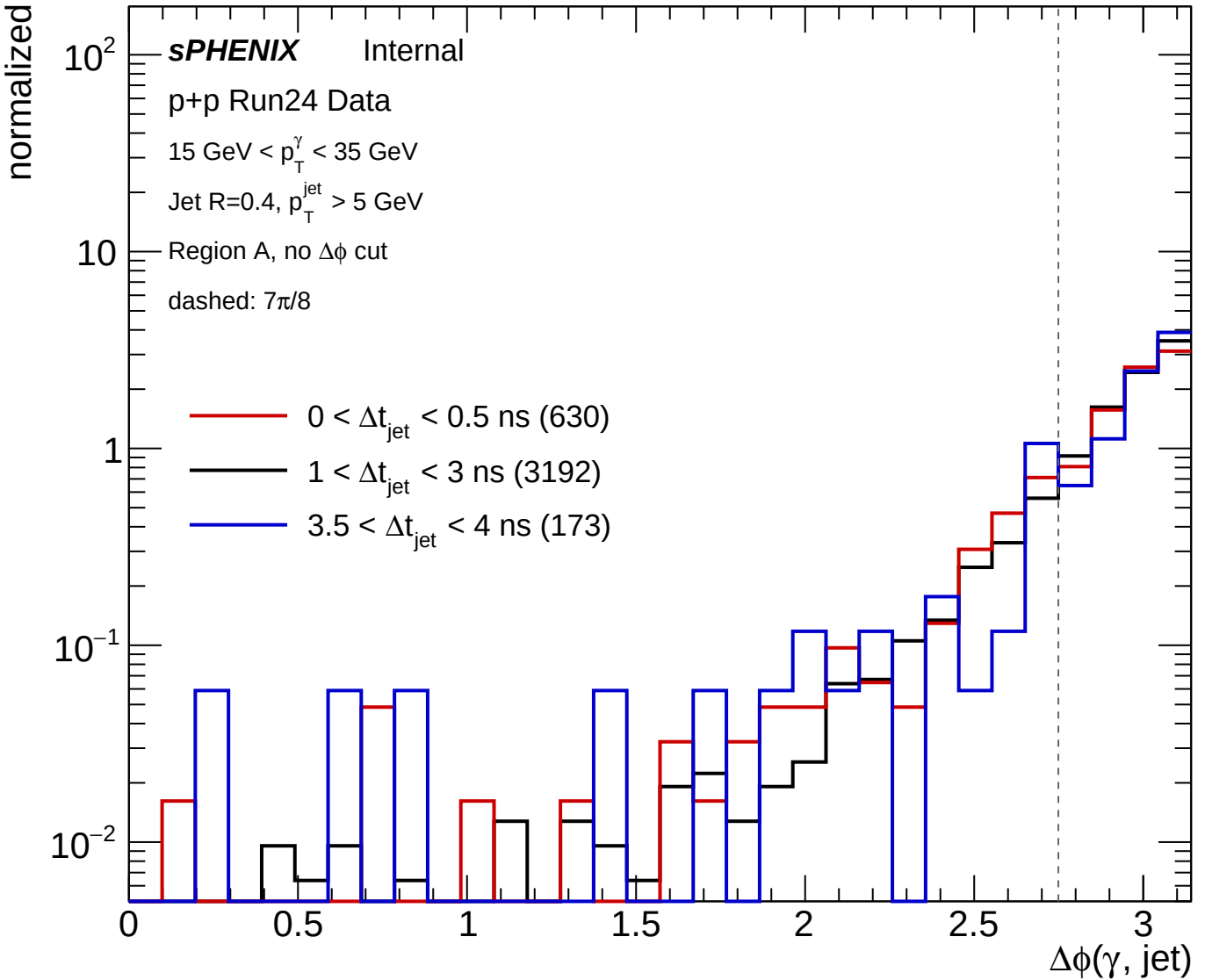

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Regions A+B+C+D, paired
slices of corrected jet p_T
p_T^{jet} 5-8 GeV (555)
p_T^{jet} 8-11 GeV (1477)
p_T^{jet} 11-14 GeV (1839)
p_T^{jet} 14-18 GeV (2473)
p_T^{jet} 18-24 GeV (2500)
p_T^{jet} 24-35 GeV (1757)
normalized
t_{MBD} - t_{jet} [ns]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
–1
0
1
2
3
4
5

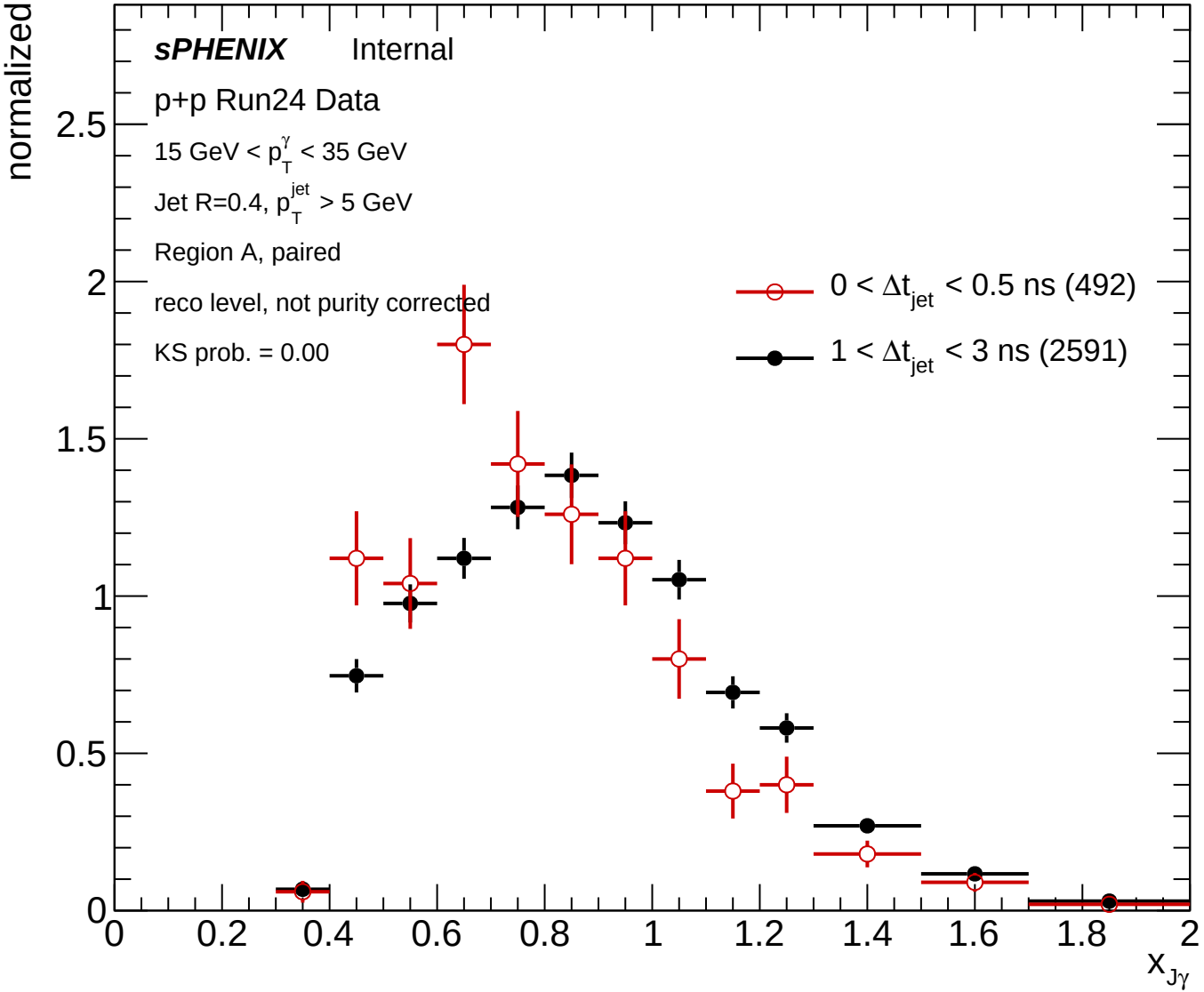

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Regions A+B+C+D, paired
slices of jet EM fraction
EM frac. 0.00-0.30 (1199)
EM frac. 0.30-0.50 (2080)
EM frac. 0.50-0.70 (2993)
EM frac. 0.70-0.85 (2424)
EM frac. 0.85-1.00 (2130)
normalized
t_{MBD} - t_{jet} [ns]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
–1
0
1
2
3
4
5

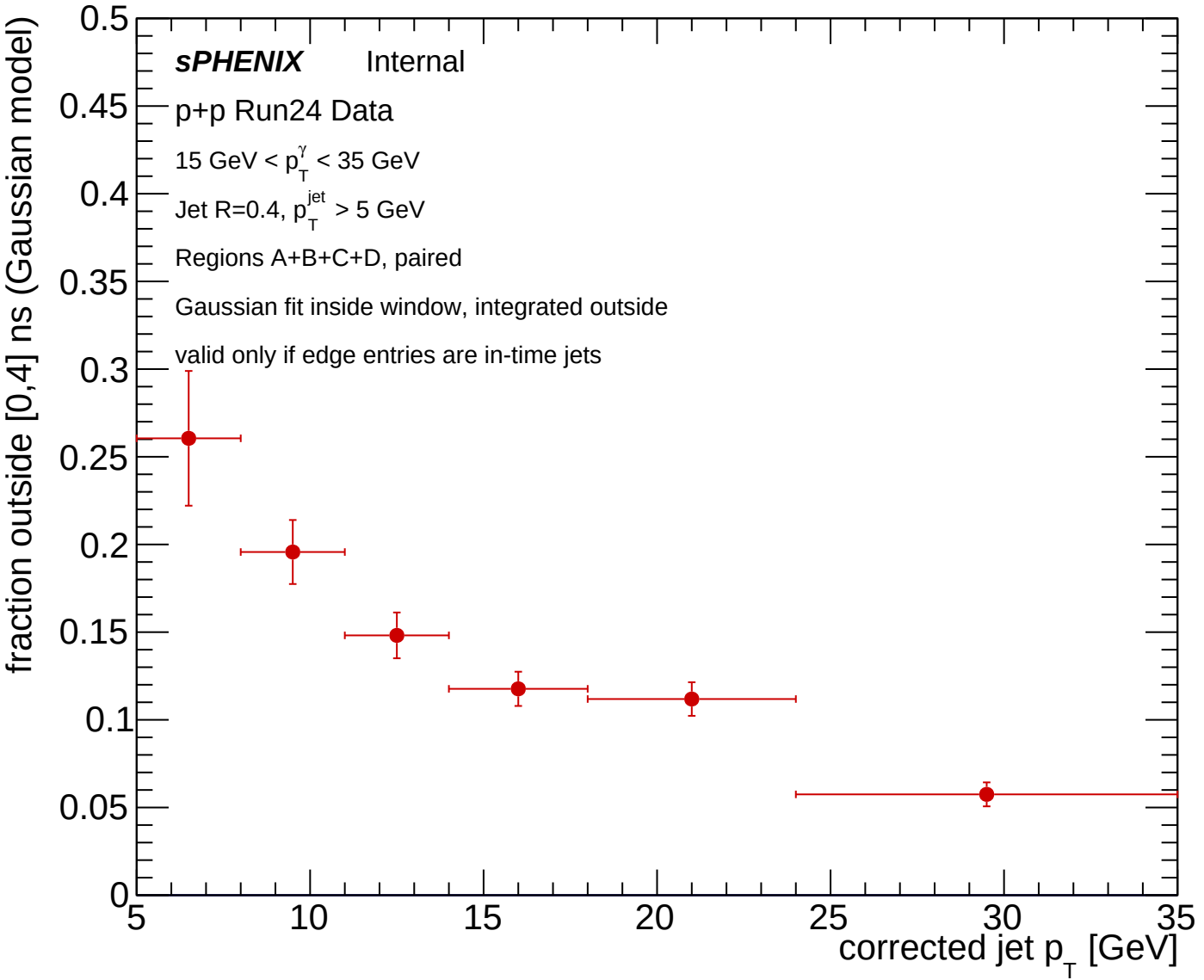

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Regions A+B+C+D, paired
dashed: window center (2 ns)
mean of in-window entries only
$\langle t_{MBD} - t_{jet} \rangle$ [ns]
corrected jet p_T [GeV]
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
5
10
15
20
25
30
35

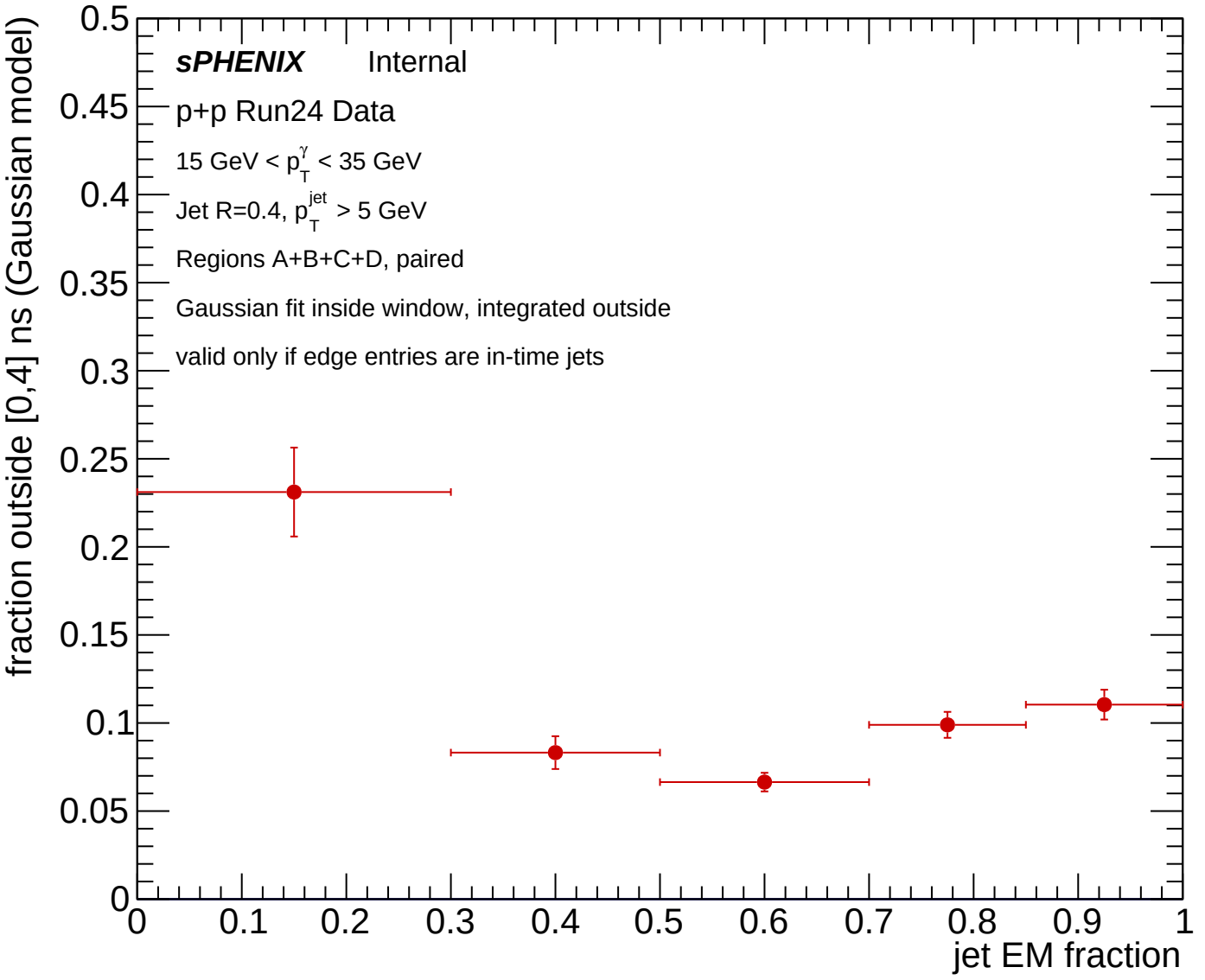

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Regions A+B+C+D, paired
dashed: window center (2 ns)
mean of in-window entries only
$\langle t_{MBD} - t_{jet} \rangle$ [ns]
jet EM fraction
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

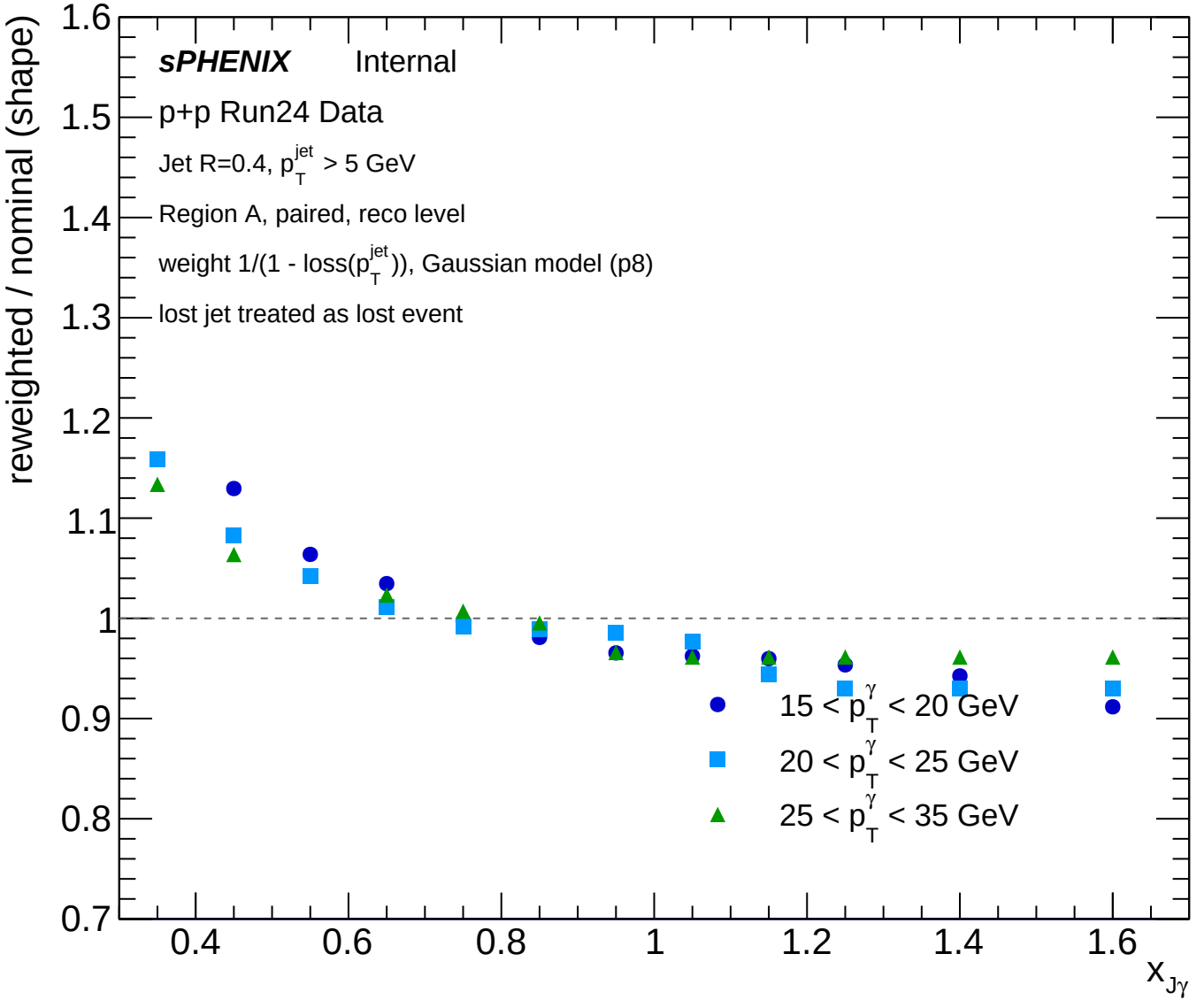

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^{γ} < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Region A, paired
corr. = 0.28
$t_{MBD} - t_{jet}$ [ns]
$t_{MBD} - t_{cluster}$ [ns]
5
4
3
2
1
0
−1
−1
0
1
2
3
4
5
25
20
15
10
5
0


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Region A, no Δϕ cut
dashed: 7π/8
0 < Δt_{jet} < 0.5 ns (630)
1 < Δt_{jet} < 3 ns (3192)
3.5 < Δt_{jet} < 4 ns (173)
normalized
10^2
10
1
10^{-1}
10^{-2}
0
0.5
1
1.5
2
2.5
3
Δϕ(γ, jet)


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Region A, paired
reco level, not purity corrected
KS prob. = 0.00
0 < Δt_{jet} < 0.5 ns (492)
1 < Δt_{jet} < 3 ns (2591)
normalized
$x_{J\gamma}$
0
0.5
1
1.5
2
2.5
0.2
0.4
0.6
0.8
1.2
1.4
1.6
1.8


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Regions A+B+C+D, paired
Gaussian fit inside window, integrated outside
valid only if edge entries are in-time jets
fraction outside [0,4] ns (Gaussian model)
corrected jet p_T [GeV]
0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
5
10
15
20
25
30
35


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^{γ} < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Regions A+B+C+D, paired
Gaussian fit inside window, integrated outside
valid only if edge entries are in-time jets
fraction outside [0,4] ns (Gaussian model)
jet EM fraction
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1


sPHENIX Internal
p+p Run24 Data
Jet R=0.4, p_T^{jet} > 5 GeV
Region A, paired, reco level
weight 1/(1 - loss(p_T^{jet})), Gaussian model (p8)
lost jet treated as lost event
15 < p_T^{γ} < 20 GeV
20 < p_T^{γ} < 25 GeV
25 < p_T^{γ} < 35 GeV
reweighted / nominal (shape)
$x_{J\gamma}$
1.6
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.4
0.6
0.8
1
1.2
1.4
1.6